3	**(a)**	Internal energy is the sum of microscopic kinetic energy and potential energy of all the molecules in the gas	**1**
		Microscopic potential energy for a real gas is not equal to zero as there is intermolecular bonding between the molecules. So internal energy is not only dependent on temperature.	**1**
	(b)	$p = \frac{1}{3}\rho\langle c^2\rangle = \frac{1}{3}\frac{M}{V}\langle c^2\rangle$, where M = total mass of the ideal gas V = total volume of the ideal gas	**1**
		$\frac{3}{2}pV = \frac{1}{2}M\langle c^2\rangle$ $\frac{3}{2}nRT = \frac{1}{2}M\langle c^2\rangle$ = Total KE of gas	**1**
		Since internal energy, U = total KE of ideal gas, as PE of ideal gas = 0 $\therefore U \propto T$	**1**
4	**(a) (i)**	$pV = NkT$	